

Arthur Hafliger

Mechanical Engineering student focused on machine design, manufacturing, precision engineering

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In depth Descriptions on [ArthurHafliger.github.io](https://github.com/ArthurHafliger)

Education - University of Central Florida - BSME (3.8 GPA), Math/Robotic Systems Minor - 2028
Pasco-Hernando State College - AA - 2025

Technical Skills

CAD	Programming	Simulation	Robotics	Manufacturing	Productivity
Solidworks(CSWP) Onshape	C++ Arduino	MATLAB	F/I kinematics PID Control	3D Printing	Jira MS Office

Relevant Coursework

Dynamics,Solid Mechanics, Principles of Electrical Engineering, Differential Equations, Engineering Concepts and Methods, Intro to C

Project Experience

Custom Servo/Cycloidal Drive - Self Directed 3/1/26 - Ongoing

Solo Project

- Designed and manufactured a 3D-printable cycloidal drive (9:1 reduction) to address limitations in torque amplification for compact robotic systems
- Achieved an operational torque value of 0.546 lbFt and a time to convergence of 1.01 seconds within a 0.5 degree resolution
- Utilized MATLAB to simulate dynamics of a motor to implement a PID controller to control a PWM Output Signal
- Optimized PID control parameters in MATLAB simulation, reducing angle convergence time by 20% relative to design requirement
- Utilized MATLAB to create graphs to monitor critical data and ensure correctness of simulations

Miniature Boat design - UCF, Orlando, Florida, 1/1/26 - 5/1/26

Team Project

- Designed and developed a 3D printed Boat Hull capable of supporting a load with a factor of safety of 7 while reducing total length to 40% of the budgeted length
- Oversaw electronic and hardware integration to Hull design reducing support material required for manufacturing and lowering total manufacturing spending by 20%
- Utilized my CSWP skills to help fellow teammates with CAD concepts such as part creation, assembly, and manufacturing

3 DOF Robotic Arm - Self Directed 12/1/25 - 2/1/26

Solo Project

- Studied robot kinematics to develop forward and inverse kinematics using a damped least squares Jacobian solver
- Implemented PID velocity control for manipulation by any 3 dimensional vector
- Studied Linear Algebra beyond what was taught in class to design a singularity detection, avoidance, and handling algorithm
- Developed 3D printer print head assembly and disassembly skills to fix a near project ending manufacturing mistake

Leadership Experience

SolidWorks Student Leadership Program - University of Central Florida, Orlando, Florida, 9/1/25 - Ongoing

Member

- Achieved completion of CSWP prep Program, scoring 13% higher than necessary on the CSWP exam
- Collaborated with peers to solve complex CAD and design challenges, improving group problem-solving efficiency
- Utilize CSWP certification to teach non-members the basics of CAD in part creation and assembly

Work Experience

Wawa - Port Richey/Oviedo, Florida, 12/9/23 - Ongoing

Customer Service Associate

- Communicate directly with customers in a fast paced and high stress environment while staying kind and collected, increasing customer satisfaction and decreasing waste of product while handling anywhere from 30-60 orders an hour
- Assume food and beverage management responsibilities when necessary to adapt to employee schedules and reduce spoilage while maintaining customer satisfaction
- Communicate directly with vendors when necessary, increasing store organization and reducing potential spoilage

Interests

